Experimental validation of a causal local corrector on top of a variational nominal trajectory for collision avoidance among moving ground robots

Slimane Aouanouk [1,2*], Mathis Benchikh [1,2], Amine Ammar [3,2], Sebastian Rodriguez Iturra [1,2], Francisco Chinesta [1,2], Dominique Baillargeat [2]

[1]PIMM Laboratory & ESI Group Chair, Arts et Metiers Institute of Technology, CNRS, Cnam, HESAM Université, 151 boulevard de l'Hôpital, Paris, 75013, France.
[2]CNRS@CREATE LTD, 1 Create Way, #08-01 CREATE Tower, 138602, Singapore.
[3]LAMPA Laboratory & ESI Group Chair, Arts et Metiers Institute of Technology, 2 boulevard du Ronceray, BP 93525, Angers cedex 01, 49035, France.

*Corresponding author(s). E-mail(s): slimane.aouanouk@ensam.eu;

Abstract

Safe navigation among independently moving agents requires global progress and reaction without access to their future motion. We evaluate an architecture that separates a variational nominal trajectory, computed offline from the static scene, from a causal local corrector driven online by observed agent states. Two independent studies comprise a simulation comparison of twelve separately calibrated nominal-generator/corrector combinations and an experimental evaluation of one combination on an encoderless differential-drive rover using overhead vision as its sole feedback. At a 10 cm terminal tolerance, the two leading variational combinations achieved joint safety-and-arrival frequencies of 65.38 % and 63.46 %, versus 48.08 % for the best straight-line combination. In forty physical trials with one or two moving agents, the rover reached the goal every time, without contact or negative clearance; the smallest agent margin was 6.3 cm, and the loop operated near 20 Hz. Paired simulation reproduced path geometry to about 5 cm mean departure but not path length. These results support feasibility in the tested configurations, without establishing a formal separation guarantee or general performance superiority.

Keywords: Trajectory optimization, Collision avoidance, Differential-drive robots, Causal control, Experimental validation, Vision-based control

1 Introduction

Ground robots increasingly share their workspace with people, handling equipment and other robots that do not announce their future motion. The robot must therefore plan a route over the full mission while reacting quickly to an agent absent from the planned scene. This separation is harder on a differential-drive rover, which cannot instantaneously execute an arbitrary velocity direction (Siegwart, Nourbakhsh, & Scaramuzza, 2011). It must also be tested on hardware, where slip, actuator dynamics, latency and perception noise are not fully reproduced in simulation (Wilson & Egerstedt, 2023).

Continuous trajectory optimisation addresses the long-horizon part of this problem. Direct methods transcribe the motion into a finite-dimensional problem (Betts, 1998); examples include covariant gradient refinement (Ratliff, Zucker, Bagnell, & Srinivasa, 2009), stochastic perturbation (Kalakrishnan, Chitta, Theodorou, Pastor, & Schaal, 2011), and sequential convex approximation with continuous collision checking (Schulman et al., 2014). The variational formulation used here follows a hybrid model-based and data-driven treatment that moves part of the computation offline (Ghnatios et al., 2024). Such a plan still depends on the scene supplied to the optimiser and cannot react to an agent excluded from that scene.

We call a component *causal* when it accesses only current and past agent states. A prediction built from past observations or a receding-horizon controller can therefore be causal. In contrast, reading an agent's scripted future positions from a simulation is not admissible. No component in the reported architecture receives an exact future agent trajectory.

Several method families address short-range avoidance. Attractive and repulsive fields provide a fast correction (Khatib, 1986), but can produce local minima, oscillation and loss of progression (Koren & Borenstein, 1991). Navigation functions remove spurious minima under structured assumptions that do not transfer directly to empirical moving-agent fields (Rimon & Koditschek, 1992). Directional histograms and dynamic windows sample local commands and can account for motion limits (Borenstein & Koren, 1991; Fox, Burgard, & Thrun, 1997). Velocity obstacles add geometric anticipation (Fiorini & Shiller, 1998); reciprocal variants distribute the avoidance effort and therefore assume some cooperation (van den Berg, Guy, Lin, & Manocha, 2011; van den Berg, Lin, & Manocha, 2008). Receding-horizon control can include non-holonomic kinematics, saturation and clearance constraints (Mao, Gao, & Guo, 2020; Zhang, Gao, & Zhang, 2020), but depends on prediction quality and the available online budget.

Combining a global reference with a local corrector is not new. Reciprocal avoidance has been applied to non-holonomic multi-robot systems (Alonso-Mora, Breitenmoser, Rufli, Beardsley, & Siegwart, 2013), and global preferred motion has been corrected under short-horizon non-holonomic constraints (Alonso-Mora, Beardsley, & Siegwart, 2018). Related work combines potential fields with a local velocity-space planner (Lin, Yue, Chen, & Sun, 2022), estimates unknown agent cooperation (Martinez-Baselga et al., 2025), or addresses cooperative and non-cooperative agents with causal state estimation on differential-drive robots (Martinez-Baselga, Lanaspa, Riazuelo, & Montano, 2026). Offline computation with external visual localisation has also been demonstrated on an omnidirectional robot in an essentially static scene (Montés et al., 2021). The less documented case studied here combines an offline variational nominal built from the static scene, a causal correction from currently observed non-cooperative agents, and an encoderless differential-drive rover using overhead vision as its only feedback.

Two conditions are central to this evaluation. Candidate architectures must be tuned independently because their ranking depends on both their parameters and the surrounding pipeline (Liu & Liu, 2022; Moll, Chamzas, Kingston, & Kavraki, 2021). Moreover, success must require both safety and arrival: a robot that stops does not collide, while a robot that arrives may still violate its clearance (Xu, Zhang, Cai, & Liu, 2023). These empirical criteria do not provide a formal guarantee, which would require an explicit model, assumptions and proof (Mitsch, Ghorbal, Vogelbacher, & Platzer, 2017).

Three hypotheses are tested. **H1**, in simulation, independently tuned nominal–corrector pairs differ in joint safety-and-arrival frequency, and their ordering is not captured by safety alone. **H2**, also in simulation, a global plan improves joint success over its ablation by an amount that depends on the terminal tolerance. **H3**, on hardware, the complete perception–decision–command chain reaches the goal without physical contact in repeated one-agent and two-agent tasks while sustaining the requested loop rate. The third hypothesis tests feasibility only in the reported domain and does not validate the other simulated architectures.

The contributions are limited accordingly. The causal separation is specified and enforced by construction. Twelve independently calibrated nominal-generator/corrector combinations

are compared using joint safety and arrival. Separately, one hardware configuration is evaluated end to end over forty runs of two frozen tasks. All runs reached the goal without contact or negative clearance, and the median loop rate was at least 19.84 Hz. Finally, paired simulation is assessed against the measured paths: it reproduces their geometry to about 5 cm mean departure but not their length. All quantitative claims remain limited to the reported campaign and testbed, with at most two moving agents.

2 Problem statement and system overview

2.1 Motion task and success

A controlled robot travels from $\mathbf{x}_0 \in \mathbb{R}^2$ to $\mathbf{x}_g \in \mathbb{R}^2$ in a bounded arena containing obstacles with arbitrary cross-sections, conservatively modeled as circular, and independently driven robots (Fig. 1). For robot and object centres $\mathbf{x}_i, \mathbf{x}_j$ and conservative radii r_i, r_j, edge-to-edge clearance is

$$d_{ij}(t) = \|\mathbf{x}_i(t) - \mathbf{x}_j(t)\| - r_i - r_j. \tag{1}$$

Clearance is measured edge-to-edge between the conservative bounding discs used by the controller (0.13 m radius per robot), not between the physical chassis outlines. A negative value therefore means that the two safety discs overlapped, not that the robots made contact. No physical contact was observed during the campaign. A run is safe when no collision, arena exit or execution error occurs, and it arrives when its final distance is no greater than tolerance τ. The success criterion is their conjunction:

$$\begin{aligned} I_s &= \begin{cases} 1, & \text{if no collision, no arena exit} \\ & \text{and no execution error occurred,} \\ 0, & \text{otherwise,} \end{cases} \\ I_g &= \mathbf{1}[\|\mathbf{x}(t_f) - \mathbf{x}_g\| \le \tau], \\ I_j &= I_s I_g. \end{aligned} \tag{2}$$

Neither safety without progress nor arrival after a margin violation is a success.

For fixed obstacles and moving agents, $d(t)$ is the minimum clearance over the objects selected for that run, and a negative value constitutes a collision for the operational indicator. This indicator is deliberately stricter than physical contact: overlap of conservative discs is sufficient for failure even when the chassis outlines do not touch. Execution errors also make $I_s = 0$, even if no negative clearance was recorded. Conversely, a positive measured clearance is an empirical observation subject to localisation error, not a proof that future executions remain separated.

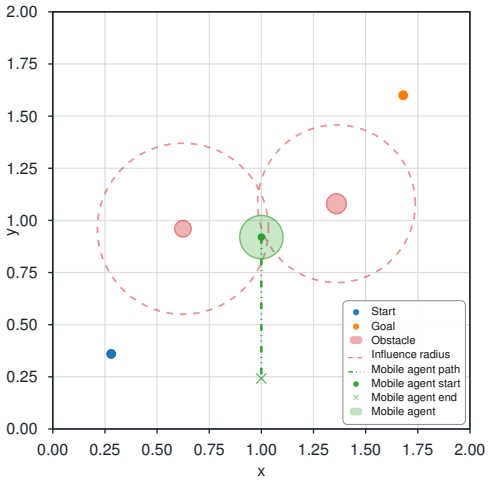


Fig. 1 Task geometry. Dashed discs define the edge-to-edge clearance used in Eq. (1). Moving agents are independently driven and are not included in the offline nominal scene.

2.2 Causality and processing chain

Here a component is *causal* when it accesses only current and past states of moving agents; their exact future state is never queried. This definition still allows limited anticipation, such as looking ahead along the nominal trajectory, predicting the controlled robot's own state, or using a short history of past observations. However, the system never uses the true future positions of moving agents. Moving agents are therefore excluded from the offline planning stage, and the local corrector relies only on their currently measured positions, within a limited perception range, optionally supplemented by a short recent observation history. It does not imply a complete sensing model or a safety proof.

Figure 2 separates seven stages: static scene, offline nominal, current observation, causal correction, wheel command, execution and measurement. The last four form the physical closed loop;

simulation, execution and measurement remain distinct throughout the analysis.

3 Materials and methods

3.1 Experimental platform and localisation

The usable metric arena is 2.00 m square within a 2.50 m printed mat. The square-section obstacles are passive, 3D-printed parts. Four commercially available four-wheel differential-drive rovers have no wheel encoders. One carries a Raspberry Pi 5 that performs vision, correction, logging and serial motor commands; the other rovers receive independent pre-programmed commands and do not react to the controlled robot. An overhead Basler ace 2 camera is the sole feedback source (Fig. 3). Frames are acquired at 1500 × 1200 Mono8 and processed near 20 Hz.

The controlled rover carries its computation and power supply; only the overhead measurement is external. The camera stream reaches the onboard computer through the bench network, whereas each moving agent receives its own trajectory and speed setpoints over an independent HTTP link. Agents are placed manually and commanded open-loop along straight lines. They neither observe the controlled rover nor share avoidance responsibility with it. Their realised trajectories are recorded and used in the physical-run analysis: over the 60 agent trajectories recorded in the 40 BIS runs, the mean ratio of net displacement to travelled distance is 0.966, documenting the drift of these encoderless platforms rather than assuming ideal straight-line disturbances.

ArUco poses (Garrido-Jurado, Muñoz-Salinas, Madrid-Cuevas, & Marín-Jiménez, 2014) are corrected for lens distortion, mapped to metres by a planar homography (Hartley & Zisserman, 2004), corrected radially for marker height, and rotated by the identified forward-axis offset. Twenty intrinsic-calibration views gave 0.191 px RMS reprojection error. Mean ground-position error after distortion correction was 5.8 mm; for the raised robot marker, parallax correction reduced mean error from 60.8 mm to 11.6 mm, with 23.1 mm maximum. These errors are smaller than the minimum reported moving-agent margin but remain part of the measurement uncertainty. In particular, the worst localisation error is more than one third of the smallest moving-agent clearance reported below, so clearance values are measurements with uncertainty rather than exact physical separations. The controlled marker was nevertheless detected in every acquired frame of the BIS campaign. Full calibration diagnostics are in Online Resource 5.

3.2 Offline variational nominal

The inherited planner operates only on the static scene. Thirty intervals span a 5 s horizon; the endpoints are fixed and the intermediate nodes minimise the midpoint-discretised action

$$J_h = \sum_{k=0}^{N-1} \Delta t \left[\frac{\alpha}{2} \left\| \frac{\mathbf{x}_{k+1} - \mathbf{x}_k}{\Delta t} \right\|^2 + V\left(\frac{\mathbf{x}_{k+1} + \mathbf{x}_k}{2} \right) \right] + \lambda_s J_s + J_v + J_a. \tag{3}$$

Here J_s penalises discrete curvature, while J_v and J_a are the soft velocity- and acceleration-bound penalties. The static potential combines goal attraction and compact-support polynomial obstacle repulsion,

$$\begin{aligned} V(\mathbf{x}) &= \tfrac{\beta}{2} \left\| \mathbf{x} - \mathbf{x}_{\mathrm{g}} \right\|^2 + \sum_k V_{o,k}(\mathbf{x}) \\ &\quad + V_w(\mathbf{x}), \\ V_{o,k}(\mathbf{x}) &= \gamma g_{o,k} \left[1 - \left(\frac{d_k(\mathbf{x})}{R} \right)^2 \right]^p, \\ &\qquad \text{if } 0 < d_k(\mathbf{x}) < R, \\ V_{o,k}(\mathbf{x}) &= 0, \qquad \text{if } d_k(\mathbf{x}) \geq R, \end{aligned} \tag{4}$$

where $g_{o,k} = 1$ for every reported obstacle and $p = 3$. This polynomial is C^2 at the support boundary and replaces the earlier inverse-distance barrier. Candidate penetrations use a finite core penalty; its formula and gradient, together with the complete nominal parameters and the definitions of J_s, J_v, J_a, are given in Online Resource 5. L-BFGS-B uses an analytic gradient (Byrd, Lu, Nocedal, & Zhu, 1995; Virtanen et al., 2020). These penalties are soft; the nominal does not provide a formal collision or kinematic guarantee. The trajectory is recomputed from the archived scene by the real-time and paired-simulation pipelines

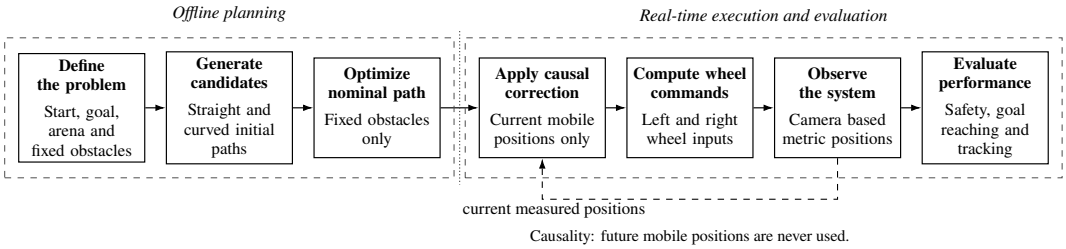


Fig. 2 Global–local architecture and offline/online boundary. Only current and retained past observations of moving agents enter the local corrector.

Table 1 Configuration used throughout the reported BIS campaign.

Group	Parameter	Symbol	Value
Geometry	Robot / fixed-obstacle radius	r_{rob}/r_o	0.130 m / 0.10 m
Hardware	Complete controlled-rover mass	m	1.258 78 kg
Corrector (hard)	Speed / acceleration bounds	$v_{\max}/a_{\max}$	0.25 m/s / 1.0 m/s^2
Corrector	Attraction / damping	k_a/c_d	5.0 / 1.4
Corrector	Mobile / static / boundary repulsion	$k_m/k_s/k_b$	0.08 / 0.03 / 0.02
Corrector	Perception radius / look-ahead	r_p/t_ℓ	0.60 m / 0.8 s
Nominal	Goal / obstacle weights	β/γ	0.20 / 5.0
Nominal (soft)	Velocity / acceleration weights	λ_v/λ_a	1.0 / 1.0
Wheels	Linear / angular inverse gains	k_v/k_ω	0.943 / 0.806
Wheels	Command range / smoothing	$u^{\max}/\alpha_{cmd}$	0.50 / 0.60
Reverse	Entry / exit thresholds	—	1.6058 / 1.6052 rad
Timing	Control period / terminal tolerance	$\Delta t_c/\tau_p$	0.05 s / 0.08 m

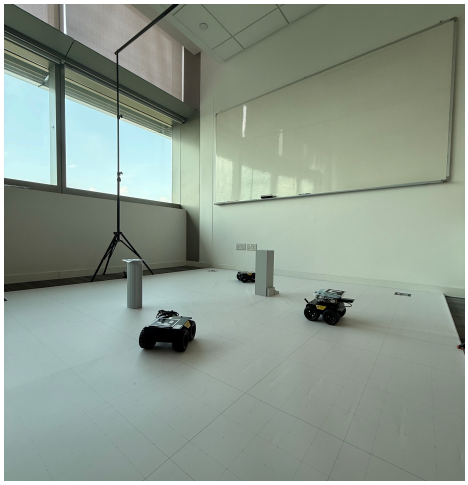


Fig. 3 Physical testbed with controlled rover, independently moving rovers, fixed square-section obstacles and overhead vision.

using the same nominal parameters; their geometric difference is not available because the original pairing convention was not archived.

A parametric multi-start family supplies the initial path, after which the bound-constrained solver performs a local optimisation. Candidate obstacle penetrations receive the finite-core penalty rather than being rejected as infeasible. Consequently, the returned nominal depends on its initialisation and weights and must not be interpreted as a globally optimal or certified path. This inherited offline stage supplies direction through the known static scene; it has no representation of the independently moving agents.

3.3 Causal local correction and wheel commands

At control time t_k, the nominal supplies a local reference and current observations supply agent positions. The force is

$$\mathbf{F}_k = k_a(\mathbf{x}_{\mathrm{ref}} - \mathbf{x}_k) - c_d\mathbf{v}_k + \sum_{j\in\mathcal{O}_k} \mathbf{F}_{j,k}^{rep} + \mathbf{F}_k^{wall}, \quad (5)$$

where $\mathcal{O}_k$ contains only currently observed objects within their influence radii. For edge distance d, unit direction $\hat{\mathbf{n}}$, gain g, influence radius H, and $d_\varepsilon = \max(d, \varepsilon)$,

$$\mathbf{F}^{rep} = g\max\left(0, \frac{1}{d_\varepsilon^2} - \frac{1}{H^2}\right)\hat{\mathbf{n}}. \quad (6)$$

The subtraction of $1/H^2$ makes the force vanish at the edge of its influence region. The moving-agent, fixed-obstacle and boundary values of H, and the numerical regularisation floor ε, are reported in Online Resource 5; all influence distances are edge-to-edge. The semi-implicit scheme used in every physical run clips acceleration and velocity in norm:

$$\begin{aligned} \mathbf{a}_k &= \mathrm{clip}(\mathbf{F}_k/m, a_{\max}), \\ \mathbf{v}_{k+1} &= \mathrm{clip}(\mathbf{v}_k + \Delta t_c \mathbf{a}_k, v_{\max}), \\ \mathbf{x}_{k+1} &= \mathbf{x}_k + \Delta t_c \mathbf{v}_{k+1}. \end{aligned} \tag{7}$$

Thus the corrector can react only to a current or retained past measurement; the look-ahead applies to the nominal reference, not to an exact future agent state. Two alternative integrators appear only in the independent simulation comparison and are documented in Online Resource 5.

This restriction is enforced directly by the system architecture. Moving agents are excluded from the offline nominal planning problem. During execution, the corrector has access only to their currently measured states and cannot query their true state at any future time. Moreover, only agents currently within the perception radius are included in the correction set. A finite observation history may support filtering or state estimation, and internal forward integration of the controlled robot remains permitted: both use information available at the decision time. What is prohibited is reading the exact scripted state of an external agent at a later instant.

The desired velocity defines heading ϕ. A feedforward/feedback yaw request is combined with hysteretic turning and reverse selection:

$$\begin{aligned} \theta &= \mathrm{wrap}(\phi - \psi), \\ \dot{\psi}_{req} &= \mathrm{sat}(\kappa_{ff}\dot{\phi}; k_\omega) + 2k_\theta k_\omega \theta, \\ \text{drive/turn thresholds} &= 1.20/0.65 \text{ rad}. \end{aligned} \tag{8}$$

Reverse entry and exit use 1.6058 and 1.6052 rad, respectively; smoothing history is cleared on direction changes. Linear and differential command components yield

$$u_L = u_v - u_\omega, \qquad u_R = u_v + u_\omega, \tag{9}$$

with joint rescaling, command floors and exponential smoothing. The hardware accepts these dimensionless wheel commands, not force or velocity. Online Resource 5 reports the complete wheel/controller parameter table and the period-rescaled smoothing law.

The mapping is intentionally hybrid. For a large heading error the rover turns in place; it enters forward or reverse translation only after the corresponding hysteresis condition is met. Joint rescaling preserves the left–right command ratio when either wheel would exceed its range. Smoothing reduces abrupt motor changes, but also adds memory and delay. These operations are executed after the planar correction and are absent from the point-mass twin, in which the rover dynamics are represented by a planar point mass with a 0.13 m conservative collision radius.

Because no encoders close the wheel loop, actuation was characterised with overhead vision. Thirty translation trials over six command levels gave

$$v = 1.5903c - 0.0888 \quad (R^2 = 0.9959), \tag{10}$$

with an extrapolated zero at $c = 0.0558$, not a measured dead-zone threshold. Seventy rotation trials showed a pronounced affine response and persistent left–right asymmetry: rightward yaw was a median 1.187 times faster. The fixed inverse gains in Table 1 are empirically tuned and are not claimed to be identified from these fits. Figure 4 retains the measured command–response evidence.

This distinction matters for interpretation. Figure 4 characterises the open-loop platform response, whereas the fixed inverse gains are the quantities used online. The affine intercept, asymmetric yaw response and lack of encoder feedback remain unmodelled residuals; characterisation does not turn the wheel command into closed-loop velocity control.

3.4 Calibration and evaluation

Four nominal generators and three correctors form twelve combinations. Each is calibrated independently in its own seven-dimensional space (Table 2); no coefficients are inherited from another combination or transferred to the physical campaign. A fixed bank contains 16 calibration and 10 held-out scenarios. Three Latin-hypercube

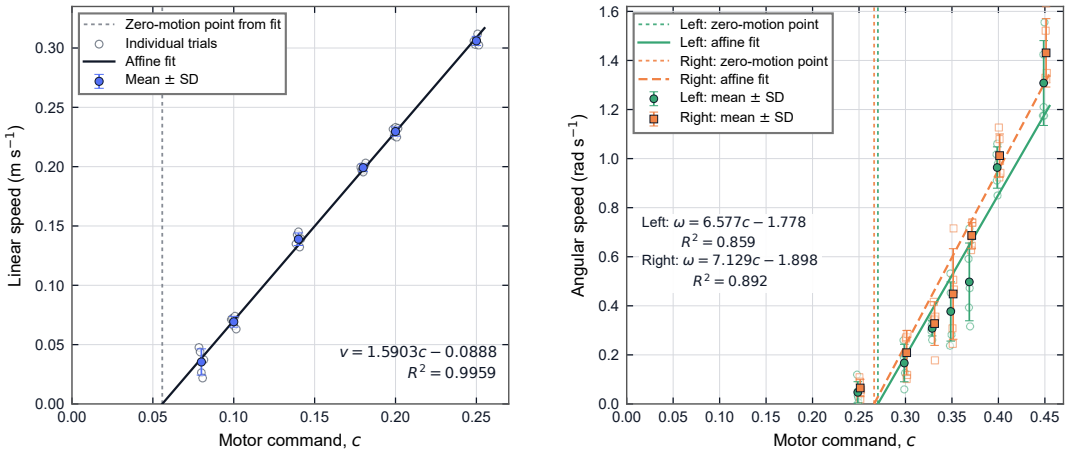


Fig. 4 Open-loop drivetrain characterisation: translation over thirty trials and rotation over seventy trials. Error bars span one standard deviation over five repetitions; zero-motion points are affine-fit extrapolations.

Table 2 Initial calibration space.

Parameter	Lower	Upper
Attraction k_a	1.50	6.50
Damping c_d	2.00	8.00
Mobile repulsion k_m	0.020	0.090
Static repulsion k_s	0.015	0.050
Boundary repulsion k_b	0.010	0.035
Perception radius r_p (m)	0.45	0.95
Look-ahead t_ℓ (s)	0.45	1.05

stages (McKay, Beckman, & Conover, 1979) contract the search region around surviving profiles. Their complete budgets, contraction rules and seven weighted score terms are given in Online Resource 5. Straight-line combinations receive the extended 900–1400–900 budget and the others the standard 400–640–400 budget; one seed is used. The unequal budgets and absence of replicated seeds limit any ranking claim.

The bank spans corridors and barriers, scattered obstacle fields, crossing traffic and an unobstructed reference, with zero to 45 static obstacles. The 16/10 split was fixed before tuning; "held out" means unused during the parameter search, not selected retrospectively by difficulty. The final common 26-scenario reevaluation therefore addresses comparability of retained profiles, but it does not remove the single-seed limitation or make their finite empirical frequencies population probabilities.

Profiles minimise a score in which any collision or arena exit dominates all soft terms:

$$S = 6000(n_m + n_s + n_b) + 100 \sum_i w_i \Pi_i. \quad (11)$$

Scenarios with less than 15 % collision-free profiles after the first stage are quarantined from that combination's score, with reinstatement thresholds of 50 % and 80 %. Consequently, scoring sets differ across combinations. Retained profiles are also reevaluated on the same 26 scenarios.

The hard counts in Eq. (11) distinguish moving-agent collision, static-obstacle collision and arena exit. The soft block combines clearance, terminal distance, acceleration saturation, commanded-force and nominal-tracking penalties. Fixed scaling makes a single hard event dominate the complete bounded soft block. Quarantine removes a scenario from a combination's primary score rather than counting it as a failure; this avoids a saturated search objective but makes the primary domains unequal by design.

Over a finite set $\mathcal{R}$ of n runs, the reported empirical frequencies are

$$R_s = \frac{100}{n} \sum_{r \in \mathcal{R}} I_s(r),$$
$$R_g = \frac{100}{n} \sum_{r \in \mathcal{R}} I_g(r;\tau), \tag{12}$$
$$R_j = \frac{100}{n} \sum_{r \in \mathcal{R}} I_j(r;\tau).$$

They are frequencies, not probabilities, and carry no confidence interval. Physical runs stop when

$$\|\mathbf{x}(t) - \mathbf{x}_g\| \le \tau_p, \qquad \tau_p = 0.08\,\mathrm{m}. \tag{13}$$

The recorded final distance is where this condition triggered, not positioning accuracy. An observed agent is classified as moving only when its net displacement exceeds 0.30 m.

For measured and simulated polylines Γ_m, Γ_s, agreement uses

$$\bar{e} = |\Gamma_m|^{-1} \sum_{\mathbf{p} \in \Gamma_m} d(\mathbf{p}, \Gamma_s), \quad e_{\max} = \max_{\mathbf{p} \in \Gamma_m} d(\mathbf{p}, \Gamma_s), \tag{14}$$

together with symmetric Hausdorff distance and path-length ratio. This is a geometric comparison; it does not measure timing agreement.

4 Results

4.1 Independent solver–controller comparison

Table 3 reports the twelve combinations at $\tau = 0.10\,\mathrm{m}$. Their scoring sets range from 6 to 14 of the 16 calibration scenarios because quarantine removes rather than penalises a scenario. The common held-out column follows reevaluation on the same 26 scenarios. The variational/Velocity–Verlet and variational/semi-implicit combinations lead the best-profile joint frequency at 65.38 % and 63.46 %; the best straight-line combination reaches 48.08 %. These values do not establish a universal ordering because budgets are unequal, only one seed is used, and the primary scoring domains differ.

Safety alone is misleading: the two no-plan combinations attain 96.15 % safety yet only 3.85 % joint success because they stall safely away from the goal (Fig. 5). The variational advantage is primarily completion rather than avoidance. It also depends on tolerance: tightening τ changes the gaps and can change the ordering. At the deliberately loose 1.0 m tolerance, for example, the straight-line/semi-implicit and leading variational combinations both reach 73.08 % joint frequency. This convergence reinforces the limited interpretation: at 0.10 m, the nominal helps precise completion in this protocol; it is not evidence of universally better avoidance. The full tolerance sweep and secondary calibration views are in Online Resource 5.

The rows also expose a second failure family. Excessively stiff repulsion can keep the bounded acceleration saturated near an obstacle and reduce both safety and progress; a global nominal does not repair such local dynamics. Thus the table is methodological context for the joint criterion, not a claim that one nominal or integrator dominates under all budgets, seeds, tolerances or scenario distributions.

4.2 Real-time budget

Across the forty physical trials, median loop rates were 19.84 Hz and 19.86 Hz for the one- and two-agent blocks. The controlled robot was detected in every acquired frame. The slowest per-run 99th percentile was 87 ms, and the longest period was 125 ms, below the configured 0.75 s guard. Table 4 shows that perception, not correction, dominates the budget. Detailed distributions and resize diagnostics are in Online Resource 5.

These loop-period statistics should not be interpreted as the total sensing-to-command delay. The control loop runs at about 20 Hz, corresponding to a nominal period of 50 ms. When a frame is processed, however, it is already about 17 ms old on average. The measured processing and serial communication stages then occupy roughly 42 ms within the control iteration before the command is emitted. The 17 ms frame age is therefore not an additional 17 ms added to the 50 ms loop period. Camera exposure, sensor readout and other uninstrumented operations are not included in these measurements. The timing results show that the implementation can sustain the requested control rate, but they do not demonstrate explicit delay compensation or provide a formal bound on closed-loop safety.

Table 3 Twelve combinations ordered by joint frequency among the four best-ranked profiles. Triples are safety/goal/joint frequencies (%) on the common 26-scenario reevaluation; the next column isolates the ten scenarios held out from calibration.

Nominal	Corrector	Common 26	Held-out R_j	Scored/16
Variational	Velocity–Verlet	76.92/69.23/**65.38**	70.00	12
Variational	Semi-implicit	80.77/72.12/63.46	70.00	13
Straight line	Semi-implicit	88.46/52.88/48.08	50.00	13
Euler–Lagrange	Velocity–Verlet	65.38/55.77/42.31	50.00	10
Euler–Lagrange	Semi-implicit	68.27/41.35/31.73	45.00	11
Straight line	Velocity–Verlet	84.62/30.77/29.81	45.00	12
Straight line	Stiff repulsion	74.04/22.12/20.19	22.50	12
Variational	Stiff repulsion	49.04/41.35/18.27	30.00	6
Euler–Lagrange	Stiff repulsion	53.85/37.50/16.35	32.50	8
No-plan ablation	Stiff repulsion	90.38/9.62/9.62	15.00	14
No-plan ablation	Semi-implicit	96.15/7.69/3.85	0.00	13
No-plan ablation	Velocity–Verlet	96.15/7.69/3.85	0.00	13

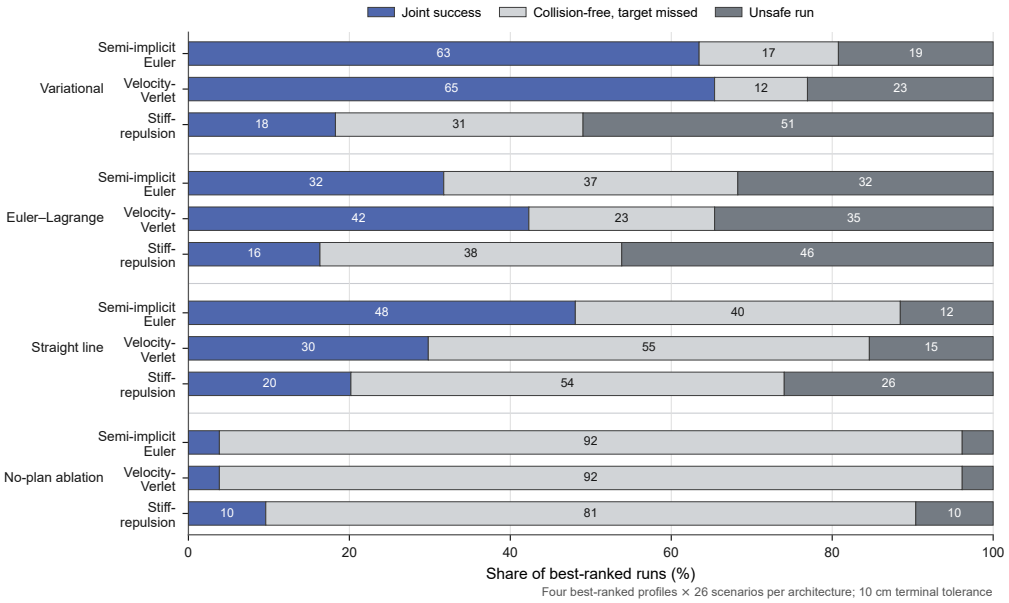


Fig. 5 Exclusive decomposition into complete success, safe but short, unsafe, and execution error. High safety can coexist with negligible joint success when the robot stalls.

Table 4 Mean time per iteration in the one-agent block.

Stage	ms	% of 50 ms
Vision	32.94	65.9
Serial transmission	8.43	16.9
Frame wait	0.83	1.7
Corrector	0.72	1.4
Logging	0.20	0.4
Remaining margin	6.88	13.8

4.3 Physical BIS campaign

The BIS campaign contains forty runs performed with one frozen coefficient set and no retuning: block A has one moving agent and 20 runs;

block B has two moving agents and 20 executed runs. Start, goal and obstacles are frozen within each block, but the blocks use different goals and fixed-obstacle layouts. The same hardware, mass, corrector, motor model, reverse thresholds, visual calibration and kinematic bounds are used throughout.

The scene is re-detected before each run rather than numerically reset. Measured starts remain within 0.7 cm of their block median, and the largest radial obstacle-position dispersion is 1.5 mm. Under the 0.30 m motion threshold, every A run contains exactly one moving agent and every B run exactly two; their measured net displacements lie between 1.02 m and 1.42 m. These checks establish the executed perimeter without treating the open-loop agent paths as identical.

One executed run, `test_B_bis_1`, retained the obstacle layout of block A because it occurred before the obstacles were moved. It reached the goal but is a valid execution of a different scene, so it is excluded from block-B aggregates solely for that reason. Results therefore use $n = 20$ for A and $n = 19$ for B while retaining the factual count of 40 executions.

All forty runs ended with `goal_reached`. No physical contact, negative conservative-disc clearance, arena exit, perception safety stop or loop-period stop occurred. The smallest moving-agent clearance was 10.6 cm in A and 6.3 cm in B. Among the runs retained in the block aggregates, the minimum static clearance was 29.7 cm; the excluded `test_B_bis_1` remained positive at 4.9 cm. These are empirical observations on two tasks, not a collision-avoidance guarantee.

Loop rate and path length are the most repeatable quantities; clearance and tracking error depend more strongly on the realised disturbance. The larger tracking error and duration in block B cannot be attributed solely to the additional agent because goal, obstacles and agent commands also differ. Figures 6 and 7 show every retained trajectory rather than a selected run.

The table supports repeatability of execution within each frozen scene, not generalisation across scenes. Final-distance dispersion is partly imposed by the stopping threshold and is not positioning accuracy. Likewise, static clearance must not be compared as an effect between A and B because the obstacle layouts differ. The stable loop-rate distributions show that adding the second observed agent did not exhaust computation in this campaign; they do not isolate its effect on the realised trajectory.

4.4 Simulation–experiment agreement

Each retained physical run was compared with a simulation initialised from pre-run measurements of the agents' positions and headings. In the simulation, agent motion was reconstructed as a constant-velocity straight-line trajectory, with speed and motion duration derived from the commanded motion and a separate drivetrain calibration. The physical and simulated runs use the same controlled-rover velocity and acceleration limits of 0.25 m/s and 1.0 m/s^2, as well as the same archived scene and nominal cost parameters. However, the nominal trajectory used in the paired simulation is recomputed independently from the archived scene and cannot be assumed to be geometrically identical to the one used during the physical run.

Consequently, differences between the agents' commanded straight-line motion and their actually realised trajectories remain one possible source of simulation–experiment disagreement. Additional discrepancies may arise because the twin applies holonomic point-mass dynamics directly, whereas the physical rover must turn, switch drive regimes, smooth wheel commands and realise motion through an affine, asymmetric drivetrain without wheel-encoder feedback. Because these effects are not isolated independently, the observed departure cannot be attributed to any single source.

Mean geometric departure is about 5 cm, but measured paths are longer (Table 6). Tenfold decimation reduces the length ratios to 1.044 and 1.128, showing that sampling noise explains part, but not all, of the discrepancy. Path length is recomputed from the pose CSV; the manifest field is cumulative and is not used as an independent measurement. The twin is a holonomic point-mass model and agreement is established only for the two repeated geometries. Online Resource 3 shows the experimental dashboard; Online Resource 5 gives the measurement-sensitivity analysis.

Table 5 Repeatability over frozen BIS tasks. Values are median [coefficient of variation]. Final distance is governed by the 8 cm stop rule.

Quantity	One agent ($n = 20$)	Two agents ($n = 19$)
Loop rate (Hz)	19.839 [0.2%]	19.863 [0.4%]
Path length (m)	1.931 [0.9%]	2.336 [3.6%]
Duration (s)	7.721 [1.1%]	10.439 [5.5%]
Static clearance (m)	0.310 [1.9%]	0.536 [3.7%]
Final distance (m)	0.073 [4.5%]	0.072 [7.4%]
Agent clearance (m)	0.117 [4.5%]	0.094 [13.0%]
Corrected commands (%)	85.2 [7.3%]	79.9 [3.6%]
Maximum tracking error (m)	0.055 [28.2%]	0.351 [12.1%]

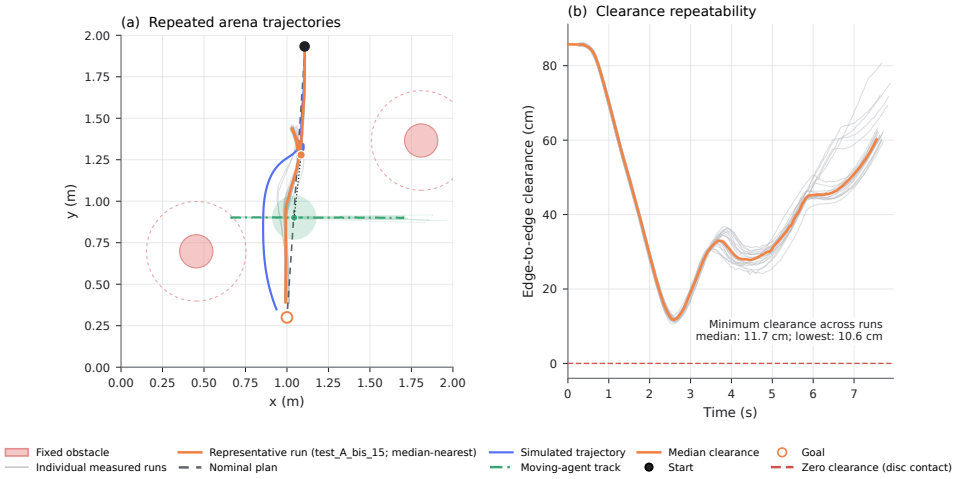


Fig. 6 Twenty physical runs with one moving agent (Online Resource 1): arena paths and conservative edge-to-edge clearance. Zero denotes overlap of the bounding discs, not chassis contact.

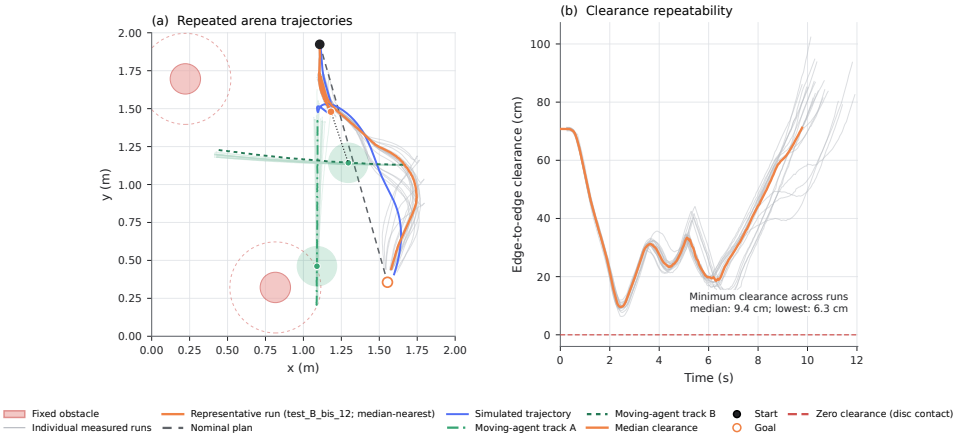


Fig. 7 Two-agent task (Online Resource 2): nineteen runs retained in the aggregates from twenty executions. Clearance is the smaller of the two agent clearances at each instant.

5 Discussion

The comparison demonstrates why safety must be paired with arrival. Local repulsion can avoid conservative-disc overlap by stopping, and local minima or force balance were observed as safe stalls. The variational nominal improves completion in this protocol, but the magnitude and even

Table 6 Median geometric agreement of paired paths.

Block	n	Mean (cm)	Haus. (cm)	Length ratio
One agent	20	5.81	14.26	1.090
Two agents	19	5.31	18.00	1.169

ordering depend on the terminal tolerance, scoring subset, unequal 900/400 budgets and single seed. These finite frequencies do not establish general superiority. The comparison and physical study answer complementary questions and neither validates the other eleven combinations on hardware.

The results support a qualitative division of roles. The offline nominal provides a route through the static scene but cannot react to an excluded agent. The online corrector reacts from current measurements but can settle into a local force balance before reaching the goal. This behaviour appears in the no-plan ablation, whereas the complete hardware chain executed avoidance manoeuvres in both tested tasks. The two studies answer different questions: the BIS campaign validates only its hardware configuration, not the ranking or alternative correctors in simulation. That hardware profile followed preliminary physical trials. Five gains predated the reported calibration campaign, and the full seven-parameter profile was finalised later rather than produced by that calibration.

The physical evidence is stronger but narrower. Forty executions of two frozen tasks reached the goal without contact or negative clearance at the requested loop rate. The correction used only 0.72 ms; vision and serial communication consumed most of the budget. Repeatability supports stable execution of these manoeuvres, not a success probability in other scenes. Soft repulsion and bounded acceleration provide no invariant separation set or formal non-collision guarantee.

Planning, simulation, execution and measurement disagree for several non-separable reasons. The twin omits non-holonomy, slip, the affine and asymmetric drivetrain, command switching, smoothing and latency. It also uses command-derived straight-line agent trajectories rather than recorded motion. The real-time and twin nominal paths were recomputed independently. Conversely, localisation noise lengthens the measured rover polyline, although decimation shows that it does not explain all of the elongation. The protocol cannot assign causal shares to these mechanisms.

The experimental perimeter is limited to one rover, at most two independently moving agents, two geometries, one day and one coefficient set. Current-state vision is treated optimistically between frames, without a delay-aware or uncertainty-aware agent model. The controller retains the structural local minima and blocking modes of potential fields. A single simulated six-agent case is provided only as an illustration (Online Resource 4, simulation only) and is not evidence of scalability.

The next physical study should vary controlled, separable factors. End-to-end timestamps would measure delay; wheel encoders and the affine response would support a non-holonomic twin; and repeated geometries across days would separate repeatability from generalisation. Causal short-horizon anticipation and global replanning could then be tested against late reactions and force-balance stalls. Comparisons with formally safe controllers should retain the same measured agent trajectories, stopping criterion and clearance definition so that differences can be attributed to the controller rather than the protocol.

6 Conclusion

An offline inherited variational nominal and a causal current-observation corrector were evaluated as a complete vision-to-wheel chain. The independent simulation study shows that joint success, rather than safety alone, exposes safe stalls and tolerance-dependent rankings. In the BIS campaign all forty executions reached the goal without contact or negative clearance, and the loop operated near 20 Hz. Paired simulation reproduced path geometry approximately but not path length. These results support feasibility and repeatability only for the tested tasks; they provide neither a formal non-collision guarantee nor evidence of general multi-agent scalability.

Supplementary information. Supplementary material accompanies the online version of this article.

Online Resource 1 (`ESM_1.mp4`): Physical single-agent run of the on-board causal controller in the instrumented arena.

Online Resource 2 (`ESM_2.mp4`): Physical two-agent crossing run using the same controller and testbed.

Online Resource 3 (`ESM_3.mp4`): Experimental workflow and dashboard, from metric tracking and paired simulation to hardware execution and trajectory comparison.

Online Resource 4 (`ESM_4.mp4`): One antipodal six-agent scenario (simulation only); it is an illustration, not a general scalability validation.

Online Resource 5 (`ESM_5.pdf`): Supporting calibration, architecture-comparison, timing and run-level BIS diagnostics.

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Data availability. Data will be made available on request.

Declaration on generative AI and AI-assisted technologies. During the preparation of this work, the authors used AI-assisted technologies (OpenAI Codex and large language models) to assist with code generation and software development, to support the drafting of the manuscript, and to proofread and improve the clarity of the article. All AI-assisted outputs were reviewed, verified, and, where necessary, corrected by the authors. The authors remain fully responsible for the scientific content, methods, results, interpretations, and final wording of the article.

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References

Alonso-Mora, J., Beardsley, P.A., Siegwart, R. (2018). Cooperative collision avoidance for nonholonomic robots. *IEEE Transactions on Robotics*, *34*(2), 404–420, https://doi.org/10.1109/TRO.2018.2793890

Alonso-Mora, J., Breitenmoser, A., Rufli, M., Beardsley, P., Siegwart, R. (2013). Optimal reciprocal collision avoidance for multiple non-holonomic robots. *Distributed autonomous robotic systems* (Vol. 83, pp. 203–216). Springer.

Betts, J.T. (1998). Survey of numerical methods for trajectory optimization. *Journal of Guidance, Control, and Dynamics*, *21*(2), 193–207, https://doi.org/10.2514/2.4231

Borenstein, J., & Koren, Y. (1991). The vector field histogram — fast obstacle avoidance for mobile robots. *IEEE Transactions on Robotics and Automation*, *7*(3), 278–288, https://doi.org/10.1109/70.88137

Byrd, R.H., Lu, P., Nocedal, J., Zhu, C. (1995). A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, *16*(5), 1190–1208, https://doi.org/10.1137/0916069

Fiorini, P., & Shiller, Z. (1998). Motion planning in dynamic environments using velocity obstacles. *The International Journal of Robotics Research*, *17*(7), 760–772, https://doi.org/10.1177/027836499801700706

Fox, D., Burgard, W., Thrun, S. (1997). The dynamic window approach to collision avoidance. *IEEE Robotics & Automation Magazine*, *4*(1), 23–33, https://doi.org/10.1109/100.580977

Garrido-Jurado, S., Muñoz-Salinas, R., Madrid-Cuevas, F.J., Marín-Jiménez, M.J. (2014). Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, *47*(6), 2280–2292, https://doi.org/10.1016/j.patcog.2014.01.005

Ghnatios, C., Di Lorenzo, D., Champaney, V., Ammar, A., Cueto, E., Chinesta, F. (2024). Optimal trajectory planning combining model-based and data-driven hybrid approaches. *Advanced Modeling and Simulation in Engineering Sciences*, *11*, 10, https://doi.org/10.1186/s40323-024-00266-2

Hartley, R., & Zisserman, A. (2004). *Multiple view geometry in computer vision* (2nd ed.). Cambridge: Cambridge University Press.

Kalakrishnan, M., Chitta, S., Theodorou, E.A., Pastor, P., Schaal, S. (2011). STOMP: Stochastic trajectory optimization for motion planning. *Proceedings of the 2011 ieee international conference on robotics and automation (icra)* (pp. 4569–4574).

Khatib, O. (1986). Real-time obstacle avoidance for manipulators and mobile robots. *The International Journal of Robotics Research*, *5*(1), 90–98, https://doi.org/10.1177/027836498600500106

Koren, Y., & Borenstein, J. (1991). Potential field methods and their inherent limitations for mobile robot navigation. *Proceedings of the 1991 ieee international conference on robotics and automation (icra)* (Vol. 2, pp. 1398–1404).

Lin, Z., Yue, M., Chen, G., Sun, J. (2022). Path planning of mobile robot with PSO-based APF and fuzzy-based DWA subject to moving obstacles. *Transactions of the Institute of Measurement and Control*, *44*(1), 121–132, https://doi.org/10.1177/01423312211024798

Liu, S., & Liu, P. (2022). Benchmarking and optimization of robot motion planning with motion planning pipeline. *The International Journal of Advanced Manufacturing Technology*, *118*, 949–961, https://doi.org/10.1007/s00170-021-07985-5

Mao, R., Gao, H., Guo, L. (2020). A novel collision-free navigation approach for multiple nonholonomic robots based on ORCA and linear MPC. *Mathematical Problems in Engineering*, *2020*, 4183427, https://doi.org/10.1155/2020/4183427

Martinez-Baselga, D., Lanaspa, D., Riazuelo, L., Montano, L. (2026). Adaptive optimal collision avoidance of dynamic agents for differential-drive robots. *Robotics*, *15*(4), 72, https://doi.org/10.3390/robotics15040072

Martinez-Baselga, D., Sebastián, E., Montijano, E., Riazuelo, L., Sagüés, C., Montano, L. (2025). AVOCADO: Adaptive optimal collision avoidance driven by opinion. *IEEE Transactions on Robotics*, *41*, 2495–2511, https://doi.org/10.1109/TRO.2025.3552350

McKay, M.D., Beckman, R.J., Conover, W.J. (1979). A comparison of three methods for selecting values of input variables in the analysis of output from a computer code.

Technometrics, *21*(2), 239–245, https://doi.org/10.2307/1268522

Mitsch, S., Ghorbal, K., Vogelbacher, D., Platzer, A. (2017). Formal verification of obstacle avoidance and navigation of ground robots. *The International Journal of Robotics Research*, *36*(12), 1312–1340, https://doi.org/10.1177/0278364917733549

Moll, M., Chamzas, C., Kingston, Z., Kavraki, L.E. (2021). HyperPlan: A framework for motion planning algorithm selection and parameter optimization. *Proceedings of the 2021 ieee/rsj international conference on intelligent robots and systems (iros)* (pp. 2511–2518).

Montés, N., Chinesta, F., Mora, M.C., Falcó, A., Hilario, L., Rosillo, N., Nadal, E. (2021). Real-time path planning based on harmonic functions under a proper generalized decomposition-based framework. *Sensors*, *21*(12), 3943, https://doi.org/10.3390/s21123943

Ratliff, N., Zucker, M., Bagnell, J.A., Srinivasa, S. (2009). CHOMP: Gradient optimization techniques for efficient motion planning. *Proceedings of the 2009 ieee international conference on robotics and automation (icra)* (pp. 489–494).

Rimon, E., & Koditschek, D.E. (1992). Exact robot navigation using artificial potential functions. *IEEE Transactions on Robotics and Automation*, *8*(5), 501–518, https://doi.org/10.1109/70.163777

Schulman, J., Duan, Y., Ho, J., Lee, A., Awwal, I., Bradlow, H., ... Abbeel, P. (2014). Motion planning with sequential convex optimization and convex collision checking. *The International Journal of Robotics Research*, *33*(9), 1251–1270, https://doi.org/10.1177/0278364914528132

Siegwart, R., Nourbakhsh, I.R., Scaramuzza, D. (2011). *Introduction to autonomous mobile robots* (2nd ed.). Cambridge, MA: MIT Press.

van den Berg, J., Guy, S.J., Lin, M., Manocha, D. (2011). Reciprocal n-body collision avoidance. *Robotics research: The 14th international symposium isrr* (Vol. 70, pp. 3–19). Springer.

van den Berg, J., Lin, M.C., Manocha, D. (2008). Reciprocal velocity obstacles for real-time multi-agent navigation. *Proceedings of the 2008 ieee international conference on robotics and automation (icra)* (pp. 1928–1935).

Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., ... SciPy 1.0 Contributors (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, *17*, 261–272, https://doi.org/10.1038/s41592-019-0686-2

Wilson, S., & Egerstedt, M. (2023). The robotarium: A remotely-accessible, multi-robot testbed for control research and education. *IEEE Open Journal of Control Systems*, *2*, 12–23, https://doi.org/10.1109/OJCSYS.2022.3231523

Xu, J., Zhang, W., Cai, J., Liu, H. (2023). SafeCrowdNav: Safety evaluation of robot crowd navigation in complex scenes. *Frontiers in Neurorobotics*, *17*, 1276519, https://doi.org/10.3389/fnbot.2023.1276519

Zhang, K., Gao, R., Zhang, J. (2020). Research on trajectory tracking and obstacle avoidance of nonholonomic mobile robots in a dynamic environment. *Robotics*, *9*(3), 74, https://doi.org/10.3390/robotics9030074